FINANCIAL MATHS

1. REVISION

In grade 9 you covered many aspects of Financial Maths which the following revision covers.

$SI = Pin$ OR $I = A - P$

SIMPLE INTEREST: $A = P(1 + in)$ **COMPOUND INTEREST:** $A = P(1+i)^n$

A = amount accumulated (final)

P = principal amount (initial)

i = interest rate written as a decimal

n = number of years

Activity 1: Simple Interest

1. Find i if, simple interest on R 1000, for 5 years is R500

$500 = 1000 \times i \times 5$

$10\% = i$

2. Find A, the future value of the following investment. R 25 000 is invested at 8% simple interest for 12 years.

$A = 25000(1 + 8\% \times 12) = R49\,000$

3. Find P if, an investment made at 12% simple interest for 3 years, gives a result of R 3400.

$3400 = P(1 + 12\% \times 3)$ $\quad P = \frac{3400}{(1+12\% \times 3)} = R2500$

4. Find the interest rate if, R 10 500 invested for 4 years, gives an amount of R 15120.

$15120 = 10500(1 + i \times 4)$ $\quad \frac{15120}{10500} - 1 = 4i$

$i = 11\%$

Activity 2: Compound Interest:

1. If R 2500 is invested at 12% compound interest for 3 years, annually, find the amount in the account at the end of 3 years.

$A = 2500(1+12\%)^3 = R3512,32$

2. An investment made at 9% compound interest for 6 years, gives an amount of R 16771. How much was initially invested?

$16771 = P(1+9\%)^6$ $\quad P = \frac{16771}{(1+9\%)^6}$

$= R\,9999,99... \approx R\,10\,000$

3. If R 25 000 is invested for 12 years, it grows to R62954.25 what is the compound interest rate?

$62954,25 = 25000(1+i)^{12}$

$\frac{62954,25}{25000} = (1+i)^{12}$

$\sqrt[12]{\frac{62954,25}{25000}} - 1 = i$

$i = 7,999...$

$i \approx 8\%$ p-a.

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